SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: **MLA0303**
COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: *Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization.(BL3,BL4,BL5)*

Assessment Tool Weightage: 40%

Dr . DUMPALA PRASANTH

QUESTIONS: 25 **Total Marks: 50**

Annexure A

Design and Develop Model

DURATION: 4 HRS (2 Sessions)

Model-Based RL Approach

1. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.

   (5 Marks)

2. Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward in Model-Based Reinforcement Learning.

   (5 Marks)

3. Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.

   (5 Marks)

4. Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.

   (5 Marks)

5. Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.

(5 Marks)

Analytic Gradient Computation

6. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.

   (5 Marks)

7. Design a concept map showing how gradients are computed and propagated during policy learning.

   (5 Marks)

8. Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.

   (5 Marks)

9. Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.

   (5 Marks)

10. Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.

    (5 Marks)

Sampling-Based Planning

11. Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

    (5 Marks)

12. Prepare a concept map showing how sampled trajectories are generated and evaluated for planning.

    (5 Marks)

13. Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy.

    (5 Marks)

14. Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.

    (5 Marks)

15. Draw a concept map comparing exhaustive search planning and sampling-based planning techniques.

    (5 Marks)

Model-Based Data Generation

16. Draw a flowchart illustrating synthetic data generation using a learned environment model.

(5 Marks)

17. Prepare a concept map showing the relationship among environment model, generated experiences, replay buffer, and policy learning.

(5 Marks)

18. Draw a block diagram explaining model-generated experience replay in Reinforcement Learning.

(5 Marks)

19. Design a flowchart showing how simulated experiences improve sample efficiency during training.

(5 Marks)

20. Draw a concept map illustrating the advantages and limitations of model-generated training data.

(5 Marks)

Value-Equivalence Prediction and Policy Optimization

21. Draw a flowchart explaining value-equivalence prediction for estimating future rewards.

(5 Marks)

22. Prepare a concept map showing the relationship among value functions, reward prediction, policy evaluation, and policy improvement.

(5 Marks)

23. Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.

(5 Marks)

24. Design a flowchart explaining iterative policy optimization using learned environment dynamics.

(5 Marks)

25. Draw a comprehensive concept map integrating Model-Based RL, Analytic Gradient Computation, Sampling-Based Planning, Model-Based Data Generation, Value-Equivalence Prediction, and Model-Based Policy Optimization into a unified Reinforcement Learning framework.

(5 Marks)

Code of Conduct:

I **_______K Uday Kiran Reddy(192425211)______** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:kuday

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Q1. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.

Model-Based Reinforcement Learning (MBRL) learns or uses a model of the environment and uses that model for planning and policy optimization.

Flowchart

START

↓

Initialize Policy and Environment Model

↓

Observe State S

↓

Select Action A using Policy π

↓

Interact with Environment

↓

Receive Reward R and Next State S′

↓

Learn / Update Environment Model

↓

Planning / Simulation using Learned Model

↓

Evaluate Candidate Actions / Trajectories

↓

Optimize Policy π

↓

Termination? ── No → Observe Next State / Repeat

└─ Yes → END

Explanation

1. The agent observes the current state.
2. It selects and executes an action.
3. The environment returns a reward and next state.
4. Experience is used to learn the environment model.
5. The learned model predicts future states and rewards.
6. Planning evaluates possible future actions.
7. The policy is optimized from planning results. The loop continues until satisfactory performance is reached.

Q2. Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward.

Flowchart

ENVIRONMENT
│ State + Action
↓
MODEL ←──── Reward information
│
│ Predicted states and rewards
↓
PLANNING MODULE
│ Candidate actions / trajectories
↓
POLICY π
│ Action
└────────────────→ ENVIRONMENT
REWARD: feedback used to evaluate actions and improve the policy

Explanation

Environment provides states and rewards. The model represents environment dynamics and reward behavior. The planning module uses the model to simulate future possibilities. The policy chooses actions. Rewards measure action quality and provide feedback for improvement. The central loop is Environment → Model → Planning → Policy → Environment.

Q3. Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.

Flowchart

START

↓

Observe Current State S

↓

Select / Explore Action A

↓

Interact with Environment

↓

Receive R and S′

↓

Store Experience (S,A,R,S′)

↓

Learn / Update Environment Model

↓

Predict Future States and Rewards

↓

Evaluate Candidate Actions

↓

Select Action with Highest Expected Return

↓

END

Explanation

The experience tuple is (S,A,R,S'), where S is the current state, A the action, R the reward, and S' the next state. The model learns transition behavior $P(S'|S,A)$ and reward behavior $R(S,A)$. The agent then uses these predictions to compare actions and select the best one.

Q4. Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.

Flowchart

ENVIRONMENT
|
↓ State, Reward
EXPERIENCE DATA (S,A,R,S')
↓
MODEL LEARNER
↓
LEARNED MODEL: $P(S'|S,A)$, R
↓
PLANNING MODULE
↓ Predicted trajectories
POLICY OPTIMIZER
↓
POLICY π
↓ Action
ENVIRONMENT

Explanation

The main components are the environment, experience collection, model learner, learned dynamics model, planning module, policy optimizer, and policy. Real experience trains the model; the model supports planning; planning results improve the policy; the policy executes actions in the environment.

Q5. Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.

Flowchart

```
REINFORCEMENT LEARNING
      |
  ┌─────┴─────┐
 ↓     ↓
MODEL-BASED   MODEL-FREE
RL          RL
  |         |
Uses / learns  No explicit
environment    environment
model          model
  |         |
Planning       Value / Policy
  |         |
Can simulate   Learns mainly
future paths   from experience
```

Explanation

Model-Based RL explicitly uses or learns an environment model and performs planning. Model-Free RL does not explicitly build an environment model; it learns value functions or a policy directly from experience.

Q6. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.

Flowchart

START

↓

Initialize Policy Parameters θ

↓

Observe State S

↓

Select Action using $\pi\theta(A|S)$

↓

Obtain Reward / Return G

↓

Calculate Objective $J(\theta)$

↓

Compute Gradient $\nabla\theta J(\theta)$

↓

Update Parameters: $\theta \leftarrow \theta + \alpha\nabla\theta J(\theta)$

↓

Converged? ── No → Repeat

└─ Yes → END

Explanation

The policy objective is the expected return, $J(\theta)=E\pi\theta[G]$. The gradient $\nabla\theta J$ indicates how parameters should change to improve the objective. For gradient ascent, $\theta new = \theta + \alpha\nabla\theta J$, where α is the learning rate.

Q7. Design a concept map showing how gradients are computed and propagated during policy learning.

Flowchart

POLICY PARAMETERS θ

↓

POLICY πθ

↓

ACTIONS

↓

ENVIRONMENT

↓

REWARDS / RETURN Gt

↓

OBJECTIVE J(θ)

↓

GRADIENT ∇θJ

↓

OPTIMIZATION ALGORITHM

↓

UPDATED PARAMETERS θ

└──────────→ IMPROVED POLICY

Explanation

A common policy-gradient expression is $\nabla_\theta J(\theta)=E_{\pi\theta}[\nabla_\theta \log \pi_\theta(a|s)\, G_t]$. Gradients quantify how policy parameters affect expected return. The optimizer uses these gradients to update parameters, improving the policy.

Q8. Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.

Flowchart

POLICY PARAMETERS θ

↓

POLICY $\pi\theta(a|s)$

↓

SELECT ACTIONS

↓

ENVIRONMENT

↓

REWARD / RETURN G

↓

OBJECTIVE $J(\theta)$

↓

GRADIENT $\nabla\theta J$

↓

PARAMETER UPDATE $\theta \leftarrow \theta + \alpha\nabla\theta J$

↓

IMPROVED POLICY

↺

Explanation

The loop aims to maximize expected reward. The environment supplies rewards, the objective summarizes performance, the gradient provides the update direction, and the optimizer changes policy parameters.

Q9. Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.

Flowchart

START

↓

Initialize Network Parameters θ

↓

Input State S

↓

Forward Pass

↓

Select Action

↓

Environment Interaction

↓

Receive Reward R

↓

Calculate Loss / Objective

↓

Backpropagate Gradients

↓

Update Parameters θ

↓

Repeat until performance converges

Explanation

Backpropagation computes gradients through the policy or value network. These gradients are propagated from the objective toward the parameters. The parameters are updated using an optimization rule, improving future decisions.

Q10. Prepare a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.

Flowchart

POLICY PARAMETERS θ

↓

POLICY πθ

↓

ACTION

↓

ENVIRONMENT

↓

REWARD

↓

OBJECTIVE / RETURN

↓

GRADIENT ∇J

↓

OPTIMIZATION ALGORITHM

↓

UPDATED θ

└──────────→ IMPROVED POLICY

Explanation

Optimization algorithms such as Gradient Ascent, Stochastic Gradient Descent, Adam, and RMSProp can use computed gradients. The overall objective is to maximize expected reward.

Q11. Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

Flowchart

START

↓

Learn / Load Environment Model

↓

Current State S

↓

Sample Candidate Action A

↓

Model Predicts Next State / Reward

↓

Generate Sampled Trajectory

↓

Calculate Return

↓

Compare Sampled Trajectories

↓

Select Best Action

↓

Execute Action in Environment

Explanation

Sampling-based planning evaluates a manageable set of sampled future trajectories rather than exhaustively exploring every possibility. The learned model predicts the consequences of sampled actions, and the highest-return option is selected.

Q12. Prepare a concept map showing how sampled trajectories are generated and evaluated for planning.

Flowchart

CURRENT STATE S
↓
LEARNED MODEL
↓
SAMPLE ACTIONS
├──→ TRAJECTORY 1 → Rewards → Return 1
├──→ TRAJECTORY 2 → Rewards → Return 2
└──→ TRAJECTORY 3 → Rewards → Return 3
↓
COMPARE RETURNS
↓
SELECT BEST PATH
↓
OPTIMAL ACTION

Explanation

For a trajectory $\tau=(s0,a0,r0,s1,a1,r1,...)$, discounted return can be estimated as $G=\Sigma\ \gamma\hat{}t\ r\_t$. The trajectory with the highest expected return is preferred.

Q13. Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy.

Flowchart

ENVIRONMENT MODEL

↓

PLANNER

↓

SAMPLED ACTIONS

A1   A2   A3   ...

↓

SIMULATE FUTURE TRAJECTORIES

↓

EVALUATE RETURNS

↓

SELECT BEST ACTION

↓

OPTIMAL POLICY

↓

ENVIRONMENT

Explanation

The planner queries the environment model, samples actions, simulates their consequences, evaluates returns, and chooses the best action. Repeated use of this process improves decision quality.

Q14. Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.

Flowchart

START

↓

Select State S

↓

Sample Action A

↓

Simulate an Episode / Trajectory

↓

Generate Return G

↓

Repeat for Many Samples

↓

Average Sampled Returns

↓

Estimate Q(S,A)

↓

Choose Action with Highest Estimated Value

↓

END

Explanation

Monte Carlo planning estimates action values from sampled complete returns. Q(s,a) can be approximated by $Q(s,a) \approx (1/N)\Sigma G_i$, where N is the number of samples and G_i is the return from sample i. More samples generally improve estimation at additional computational cost.

Q15. Draw a concept map comparing exhaustive search planning and sampling-based planning.

Flowchart

```
PLANNING METHODS
├── EXHAUSTIVE SEARCH
│   ├─ Explores all / very large set of possibilities
│   ├─ High computation
│   ├─ Potentially exact when feasible
│   └─ Poor scalability
└── SAMPLING-BASED PLANNING
    ├─ Explores selected samples
    ├─ Lower computation
    ├─ Approximate solution
    └─ Better scalability
```

Explanation

Exhaustive search is comprehensive but can become computationally infeasible as the search space grows. Sampling-based planning sacrifices exactness for efficiency and scalability.

ve learning framework.